

The empirical recurrence for OEIS A253464, proved by transfer matrix

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Abstract

OEIS A253464 counts arrays of width 4 over $\{0, 1, 2\}$ in which the largest entry on the main diagonal minus the smallest on the antidiagonal of every 2×2 subblock differs from the same number at the neighbouring subblock. The entry carries an empirical recurrence of order 4 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: a block of k rows and the block one row lower together occupy $k + 1$ consecutive rows, so every comparison the condition makes is settled inside 3 consecutive rows and the count is a walk count on $S = 3722$ states.

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1 The conjecture

OEIS A253464 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with every 2×2 subblock diagonal maximum minus antidiagonal minimum unequal to its neighbors horizontally, vertically, diagonally and antidiagonally.”. It has offset 1 and begins

3640, 1400, 2000, 2600, 5000, 7784, 17000, 28520, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 5 \cdot a(n-2) - 4 \cdot a(n-4)$ for $n > 6$.

As of the “Last modified” line on the live entry (Dec 14 2018 06:29:27, revision 12) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The arrays have 4 columns and a growing number of rows, with entries in $\{0, 1, 2\}$.

For a $k \times k$ block B of cells of the array, let

$$f(B) = \max_{1 \leq t \leq k} B_{t,t} - \min_{1 \leq t \leq k} B_{t,k+1-t},$$

which in words is the largest entry on the main diagonal minus the smallest on the antidiagonal. The condition is that f takes different values on any two blocks that are neighbours in one of the directions the entry names: writing $B(i, j)$ for the block whose top-left cell is (i, j) ,

$$f(B(i, j)) \neq f(B(i + \delta_1, j + \delta_2)) \quad \text{for } (\delta_1, \delta_2) \in \Delta,$$

whenever both blocks lie wholly inside the array, where

$$\Delta = \{ (1, -1), (1, 1), (0, 1), (1, 0) \}$$

collects the steps for the directions the entry names, namely antidiagonal, diagonal, horizontal and vertical. The requirement is imposed for that one size of block, 2×2 .

Two remarks fix the reading. A neighbouring block is the block one step over, overlapping the first, not the next disjoint block: the entry says “its neighbors”, and the counts below confirm it. And a comparison is imposed only when both blocks lie wholly inside the array, so a block near an edge is compared in fewer directions and a block whose named neighbour falls off the array is under no condition in that direction.

3 A window of 3 rows

A $k \times k$ block occupies k consecutive rows, and each of its neighbours occupies either the same k rows or the k rows one lower. So both blocks of any comparison lie inside $k + 1$ consecutive rows, hence inside 3 consecutive rows for every size the entry names. No comparison reaches further than that, in either direction.

Take the array one row at a time and let the state be the last 2 rows written so far, with a distinguished start standing for the array before its first row, so that a walk of n steps is an array of n rows. Appending a row produces a window of 3 rows, and the step is allowed exactly when every comparison whose two blocks both lie inside that window holds.

Every comparison of the finished array is tested this way. Each one lies inside 3 consecutive rows, and every 3 consecutive rows of the array form the window of some step. A comparison lying inside more than one window is tested more than once, which costs nothing. Conversely no step imposes anything not asked by the definition, since a step tests only comparisons whose two blocks lie inside the array.

Every state may end an array, so $\tau = \mathbf{1}$. A block sitting in the final rows whose neighbour would lie below the last row is not compared in that direction at all — the neighbour is not part of the array — so stopping never leaves a condition unchecked. For the same reason the array of exactly 2 rows is handled by the walks of length 2 with no further test: all of its comparisons already lie inside the windows of those steps.

Thus

$$a(n) = \iota^\top M^j \tau, \quad S = 3722,$$

for the transfer matrix M on those S states. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 1. *Let $q(t) = t^4 - \sum_i c_i t^{4-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+4} - \sum_i c_i M^{j+4-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 5a(n-2) - 4a(n-4)$$

for every $n > 6$.

Proof. The vector $w = q(M)\tau$ was formed by 4 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 6 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 26 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: every array of width 4 over $\{0, 1, 2\}$ with a few rows was written out cell by cell, every block of that one size was located, its number computed, and the comparison made with each neighbouring block that lies inside the array. No window, no state and no end vector are involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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